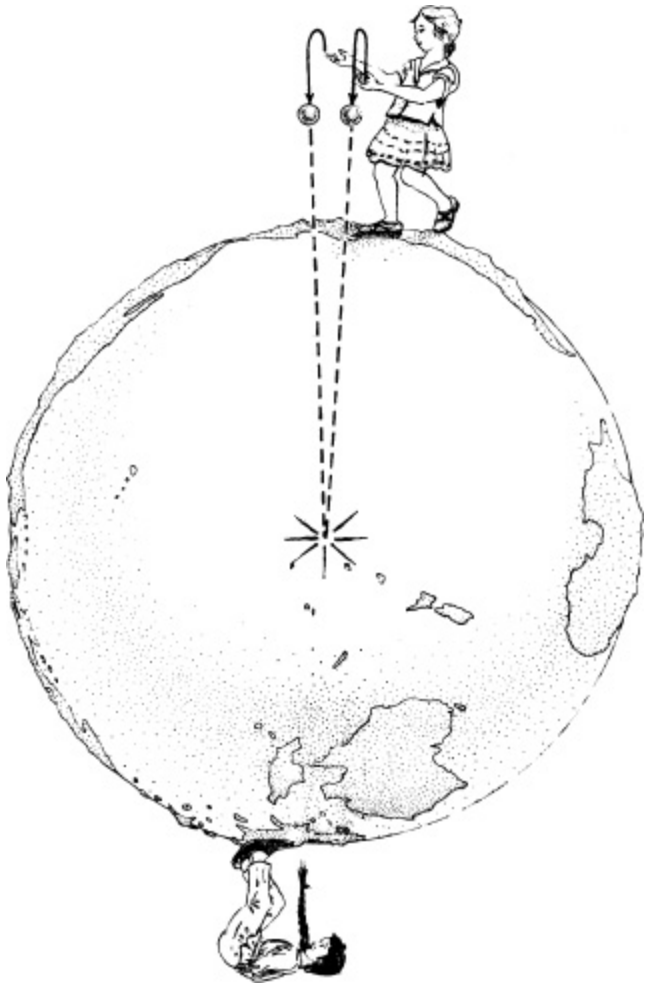


2.5 Two balls thrown into the air on precisely parallel trajectories, if able to pass unimpeded through the Earth, will collide near the Earth's center.

As the balls fall into the Earth, the Earth's tidal gravity squeezes them together in the same way as it squeezes your sides if you are a falling astronaut (Figure 2.3). The strength of the tidal gravity is just right to make both balls fall almost precisely toward the Earth's center, and hit each other there.

Now comes the payoff of this thought experiment: Each ball moved along a precisely straight line (a geodesic) through spacetime. Initially the two straight lines were parallel. Later they crossed (the balls collided). This crossing of initially parallel, straight lines signals a curvature of spacetime. From Einstein's viewpoint, spacetime curvature *causes* the crossing, that is, causes the balls' collision, just as the curvature of the globe caused straight lines to cross in Figure 2.4. From Newton's viewpoint, tidal gravity causes the crossing.